

Unit 16: Shaped by Coastal Processes

Reading: pages 226–241 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. Three factors determine how large ocean waves become. They are
 - a) wind speed, how long the wind blows, and the fetch
 - b) water temperature, salinity, and depth
 - c) tidal range, moon phase, and latitude
 - d) wavelength, amplitude, and velocity
2. The water beneath a passing wave is disturbed to a depth of approximately
 - a) twice the wavelength
 - b) the wave's amplitude
 - c) one-half of the wavelength
 - d) the full wavelength
3. Longshore drift is the combined effect of
 - a) wave refraction and coastal straightening
 - b) eustatic and isostatic sea-level change
 - c) tides and rip currents
 - d) the longshore current in the surf zone plus swash and backwash on the beach
4. An elongated sandy deposit that extends out into open water in the direction of the longshore current is called
 - a) a spit
 - b) a barrier island
 - c) a berm
 - d) a tombolo
5. Barrier islands are common along which coastline?
 - a) The Pacific coast of Canada and the northwestern United States
 - b) The U.S. Gulf Coast from Texas to Florida and the Atlantic Coast to Massachusetts
 - c) The fjord coast of Norway
 - d) Any coast that has been recently uplifted by tectonic activity

6. Groins are built perpendicular to the beach and trap sediment by
- a) blocking incoming waves entirely
 - b) slowing the longshore current
 - c) raising the local sea level
 - d) increasing the tidal range

Part B · True or False

2 questions × 1 point = 2 points

7. On an irregular coast, wave energy is focused on the headlands, which over millions of years straightens the coastline.

TRUE / FALSE

8. Beaches are typically wider and flatter in winter than in summer.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain why damming a river can put beaches many kilometers away at risk.

10. Name and explain the three mechanisms of sea-level change.

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